[1] >[aKc5Agp1q5Y] /

The Archive

[1.1] >

rish987.github.io

Introduction.

This is my personal archive, a place where I plan to record what I've learned in the fields of mathematics, computer science, and more. [1.2] >

A continual work in progress, the intent is for it to be a completely self-contained repository, with descriptions that are as concise as physically possible.

The archive is free and open-source, together with the archiver, the tool I use to build archives. You can find them below:

`github.com/rish987/archive`

`github.com/rish987/archiver`.

Navigation.

It's organized hierarchically into units called "references," which can be one of the following types: [1.3] >

- "root." The reference you're currently on! There's only one reference of this type. It's reserved for descriptions that apply to the entire archive. [1.4] >
- "topic" (denoted by T). These repeat a general format: [1.5] >
  - Describe a simple concept that relies on a simple physical intuition.
  - Correspond this intuition with some initial notation, and possibly define some more notation using this initial notation.
  - Use this notation to describe some facts that follow from it. These facts are merely stated, not proven.
- "definition" (denoted by D). These introduce notation that depend on a previously described physical intuition, previously defined notation, or a combination of both. [1.6] >
- "proof" (denoted by P). These state a fact and show why it is true, using previously defined notation. [1.7] >
- "note" (denoted by N). These may be one of the following: [1.8] >
  - A visual/verbal interpretation of a concept that isn't strictly necessary for understanding, but could be helpful.
  - A temporary stand-in for a reference that will be properly integrated into its own topic at a later time.

Each reference is delimited by a pair of horizontal rules. [1.9] >

At the upper-left corner of each reference is a "reference number" ("RN" for short), a unique ID that will never, ever change. When verbally referring to a particular reference, we say "RN" followed by the reference number, e.g. "RN53." The numbers themselves don't actually mean anything, other than roughly tracking the order in which I created the references. In fact, some reference numbers may not actually exist, if I decided to delete its former reference! [1.10] >

In the margins of a reference are labels for "reference parts" and their associated "reference part numbers," ordered starting from 1. These are logical sections of the reference that should be verbally referred with "RNP" followed by the reference part number, e.g. "RNP53.3." [1.11] >

The symbol > denotes a link to a YouTube video for that reference. It should never be strictly necessary to watch the video to understand the reference, as all the details should be in the text. [1.12] >

At the upper-right corner of each reference is a "reference path," a unique path that identifies this reference's place in the hierarchy of references. It's written in UNIX-SHELL path style. [1.13] >

A reference is linked to either explicitly or implicitly. These links may either be "local" or "global." The difference is contextual – the first link to a reference in the hierarchy is a "local" link, in that following it will append to the current reference path. All other links are "global," in that the current reference path has no relation to that of the linked reference. [1.14] >

- An explicit link has the form $[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$. [1.15] >
- An implicit link is present in the notation, and links to the reference where the notation was originally defined. Just as notation is nested, so too are the corresponding implicit links. [1.16] >

  We follow the convention that implicit links are always global; it follows that the first link to a reference is always explicit.

Annotations.

Equations may be annotated as, for example: [1.17] >

$$\overset{\texttt{i}}{a = b} \underset{\texttt{y}}{=} \overset{\texttt{j}}{c} \underset{\texttt{z}}{\geq} d,$$
$$\underset{\texttt{x}}{}$$

followed by

$$\overset{\texttt{(x-z)->}[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}}{g = h},$$

where blue underset boxes define "labels," and grey overset boxes provide "explanations" that explain why an (in)equality follows, using previously defined labels and other references. An explanation is formatted as follows:

- `x` indicates to consider the single (in)equality labeled by `x`. [1.18] >

  In the example above, `x` refers to $a = b$.

- `(x-y)` indicates that, when `x` and `y` appear in a chain (in)equality, to consider the equation that results from cropping out what appears in between the (in)equalities labeled by `x` and `y`. [1.19] >

  In the example above, `(x-z)` refers to $a \geq d$.

- `x->`$[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$ indicates to use (in)equality `x` as the antecendent of the reference linked to by $[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$. [1.20] >

  In the example above, `(x-z)->`$[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$ refers to use $a \geq d$ as an antecedent of the reference with RN `[RN]`.

Note that reference paths and labels are simply meant to be unique, memorable identifiers, and by themselves don't actually matter. Though technically, they always are acronyms/abbreviations of *something* – see if you can figure out what they are in the YouTube video comments! [1.21] >

Formats.

The archive is available in the following formats: [1.22] >

- `tree_online`: Each reference has its own online link in a "reference tree" and corresponding PDF. [1.23] >

  Links are online hyperlinks.

- `tree`: Local download of the reference tree. [1.24] >

  Links are remote PDF links.

- `full`: All references are combined into a single PDF, ordered by depth (i.e., a linearized DAG), where each reference starts on a new page. [1.25] >

  Links are cross-referencing links within the same PDF.

- `full_compact`: Identical to `full`, except references may start on the same page. [1.26] >

  Links are cross-referencing links within the same PDF.

For the best experience, I highly recommend using `pdf.js` (native to the Firefox web browser, but also available on Chrome) as your PDF viewer, in "presentation mode" (`ctrl+alt+p`) on a vertical monitor. Note that with `pdf.js`, you can use your browser's history navigation functionality to traverse your link history (`alt+rightarrow` and `alt+leftarrow`). [1.27] >

For offline viewing, I highly recommend Atril Document Viewer. [1.28] >

Topics.

[Probability Theory]$^{35}_{\mathrm{T}}$

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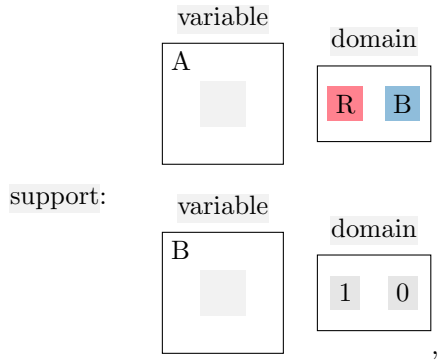
[35] /T/prob

Probability Theory

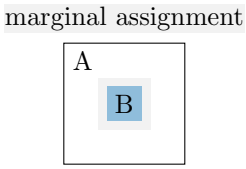
Physical intuition.

Suppose that we have a set of "variables," each with their own finite "domains", which we'll collectively refer to as the "support":



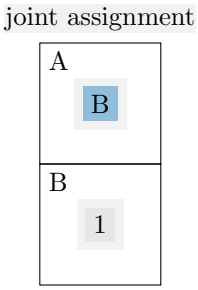
,

where a "marginal assignment" consists of associating some (but not all) of the variables of the support with something from their domains:



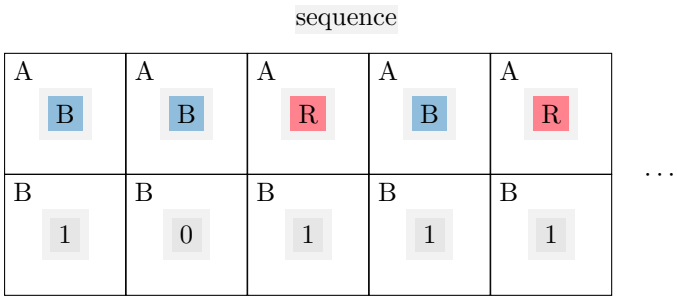
,

a "joint assignment" ("assignment" for short) consists of associating each variable of the support with something from its domain:



,

a "sequence" consists of an infinite, ordered list of such joint assignments:



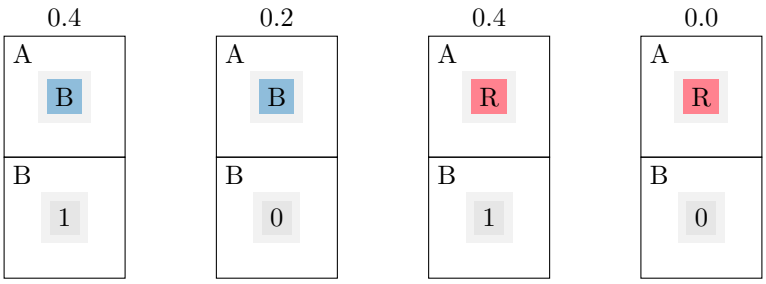
,

and a "trial" denotes a particular item (i.e. an indexed joint assignment) of this sequence.

Now, suppose something in particular about this sequence: if an joint assignment occurs at any trial, it will occur in infinitely many trials, and, additionally, as we the traverse sequence in order, the ratio of occurrences of the joint assignment to trials so far becomes arbitrarily close to some positive number. Otherwise, if an joint assignment never occurs, the ratio remains at zero. Let's call this ratio the "sequence probability" of the joint assignment. Let's refer to such a sequence as a probability sequence.

For example, if the sequence above repeated the first five trials, it would be a probability sequence defining the sequence probabilities:

sequence probabilities



Let a "distribution" be a set of probability sequences with the same sequence probability for every assignment, which, in the context of a distribution, we will simply refer to as the "joint probability" ("probability" for short) of that assignment.

Notation.

Let's correspond this intuition with some notation, and use this notation to arrive at a few more definitions.

[Define][36]

$$\mathrm{X} := \text{an arbitrary variable with domain } \mathbf{X},$$

where we use "x" to denote an element of $\mathbf{X}$.

We implicitly maintain that anything we describe using X and $\mathbf{X}$ also applies when these are replaced with one or more variables/domains: $\mathrm{X}/\mathbf{X}, \ldots, \mathrm{X}'/\mathbf{X}'$. Notational expansions are described where applicable.

Define $\mathrm{Y}, \mathbf{Y}, \mathrm{y}$, and $\mathrm{Z}, \mathbf{Z}, \mathrm{z}$ similarly.

Define $?, \mathbf{?}, ?$ similarly, but instead implicitly maintain that anything we describe using ? and **?** also applies when these are replaced with *zero* or more variables/domains.

[We let the notation][37]

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\} \text{ establish a support consisting of variable } \mathrm{X} \text{ with domain } \mathbf{X},$$

which expands as

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}.$$

[Let][56], for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, the notation

$$\mathrm{x} \in \mathbf{X}$$

indicate an arbitrary x in $\mathbf{X}$.

For contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}'), (?, \mathbf{?})\}$, this expands as

$$\mathrm{x} \in \mathbf{X}, \ldots, \mathrm{x}' \in \mathbf{X}'$$

to indicate an arbitrary combination of x in $\mathbf{X}$, ..., x′ in $\mathbf{X}'$.

[Let][57], for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$] and all $\mathrm{x} \in \mathbf{X}$,

$$\mathrm{X} = \mathrm{x}$$

denote a joint assignment (assignment for short) [or, marginal assignment (in simple English, a list of equalities that doesn't cover the entire support is a marginal assignment)].

If a joint assignment or marginal assignment was previously specified, we may later drop "X =", for "implied assignment":

$$\mathrm{x}.$$

If a joint assignment or marginal assignment was never specified, we assume an arbitrary $\mathrm{x} \in \mathbf{X}$ for "implied generality":

$$\mathrm{X}.$$

For contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}'), (\mathrm{Y}, \mathbf{Y}), \ldots, (\mathrm{Y}', \mathbf{Y}')\}$] and all $\mathrm{x} \in \mathbf{X}, \ldots, \mathrm{x}' \in \mathbf{X}'$, This expands as

$$\mathrm{X} = \mathrm{x}, \ldots, \mathrm{X}' = \mathrm{x}',$$

with implied assignment as

$$\mathrm{x}, \ldots, \mathrm{x}',$$

and with implied generality as

$$\mathrm{X}, \ldots, \mathrm{X}'.$$

[Let][38], for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQ} \text{ denote the set of all sequences},$$

where "sq" denotes an arbitrary such sequence.

Let $1 \leq t < \infty$ denote an arbitrary trial.

Because sequences are infinite (and cannot be written out), we use the notation

$$\mathrm{SQI}\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) := \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$.

[Define]$^{40}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \mathrm{SQI}\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right)$$

if such a limit exists, and undefined otherwise; that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[Let]$^{41}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}, (\exists\ 1 \leq t < \infty : \mathrm{SQI}\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) = 1) \implies \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) > 0\},$$

that is, the set of all probability sequences, where "sqp" denotes an arbitrary such probability sequence.

[Define]$^{42}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\ \mathrm{d}, \mathrm{d}' \in \mathrm{D}, \forall\ \mathrm{x} \in \mathbf{X}, \Pr\left(\mathrm{d}; \mathrm{X} = \mathrm{x}\right) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions, where "D" denotes an abitrary such distribution, and "d" denotes an arbitrary sequence in D.

[Define]$^{43}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$ and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\left(\mathrm{X} = \mathrm{x}\right)$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr\left(\mathrm{X} = \mathrm{x}\right) = \Pr\left(\mathrm{d}; \mathrm{X} = \mathrm{x}\right),$$

that is, a joint probability.

[Define]$^{46}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\left(\mathrm{X} = \mathrm{x}\right) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr\left(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right),$$

that is, a marginal probability.

[It can be shown]$^{47}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr\left(\mathrm{X} = \mathrm{x}\right) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr\left(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right).$$

[Define]$^{48}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$,

$$\Pr\left(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) = \frac{\Pr\left(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)}{\Pr\left(\mathrm{Y} = \mathrm{y}\right)},$$

and undefined when $\Pr\left(\mathrm{Y} = \mathrm{y}\right) = 0$; that is, a conditional probability.

Propositions.

[It can be shown that]$^{50}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}) = 0 \implies \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

[It can be shown that]$^{51}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}) > 0.$$

[It can be shown that]$^{52}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}, \mathrm{z} \in \mathbf{Z}$, when $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, and $\Pr(\mathrm{Y} = \mathrm{y}) > 0$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1 \implies \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 1.$$

[It can be shown that]$^{53}_{\mathrm{P}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}, \mathrm{z} \in \mathbf{Z}$, when $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, and $\Pr(\mathrm{Y} = \mathrm{y}) > 0$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 0 \implies \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0.$$

Topics.

[Reinforcement Learning Theory]$^{2}_{\mathrm{T}}$

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[53] /T/prob/P/pzerocind

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}, \mathrm{z} \in \mathbf{Z}$, when $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, and $\Pr(\mathrm{Y} = \mathrm{y}) > 0$,

$$\overset{\texttt{mg}}{} \qquad \overset{\texttt{mmg}}{}$$

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 0 \implies \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0.$$

$$\texttt{cz}$$

Proof.

We have that

$$\overset{\texttt{(mmg)->}[\ldots]^{48}_{\mathrm{D}}}{}$$

$$\Pr(\mathrm{x} \mid \mathrm{y}) = \frac{\Pr(\mathrm{x}, \mathrm{y})}{\Pr(\mathrm{y})} \overset{\texttt{cz}}{=} 0,$$

so

$$\Pr(\mathrm{x}, \mathrm{y}) = 0,$$

$$\texttt{mz}$$

and we have

$$\overset{\texttt{(mz)->}[\ldots]^{50}_{\mathrm{P}}}{}$$

$$\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z}) = 0,$$

$$\texttt{fz}$$

so

$$\overset{\texttt{(mg)->}[\ldots]^{48}_{\mathrm{D}}}{}$$

$$\Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) = \frac{\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z})}{\Pr(\mathrm{y}, \mathrm{z})} \overset{\texttt{fz}}{=} 0.$$

[52] /T/prob/P/ponecind

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}, \mathrm{z} \in \mathbf{Z}$, when $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, and $\Pr(\mathrm{Y} = \mathrm{y}) > 0$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \overset{\texttt{mg}}{=} 1 \overset{\texttt{co}}{\implies} \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \overset{\texttt{mmg}}{=} 1.$$

Proof.

We have that

$$\Pr(\mathrm{x} \mid \mathrm{y}) = \frac{\Pr(\mathrm{x}, \mathrm{y})}{\Pr(\mathrm{y})} \overset{\texttt{(mmg)->}[\ldots]^{48}_{\mathrm{D}}}{\overset{\texttt{co}}{=}} 1,$$

so

$$\Pr(\mathrm{x}, \mathrm{y}) \overset{\texttt{me}}{=} \Pr(\mathrm{y}) \overset{[\ldots]^{47}_{\mathrm{P}}}{=} \Pr(\mathrm{x}, \mathrm{y}) + \sum_{\mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x}} \Pr(\mathrm{x}', \mathrm{y}),$$

and we have

$$\forall \mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x}, \Pr(\mathrm{x}', \mathrm{y}) = 0,$$

so

$$\forall \mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x}, \Pr(\mathrm{x}', \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{50}_{\mathrm{P}}}{\underset{\texttt{az}}{=}} 0,$$

and therefore,

$$\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z}) \overset{\texttt{az}}{\underset{\texttt{arm}}{=}} \Pr(\mathrm{x}, \mathrm{y}, \mathrm{z}) + \sum_{\mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x}} \Pr(\mathrm{x}', \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{47}_{\mathrm{P}}}{\underset{\texttt{rm}}{=}} \Pr(\mathrm{y}, \mathrm{z}) \overset{\texttt{mg}}{\underset{\texttt{gf}}{>}} 0,$$

so

$$\Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) = \frac{\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z})}{\Pr(\mathrm{y}, \mathrm{z})} \overset{\texttt{(mg)->}[\ldots]^{48}_{\mathrm{D}}}{\overset{\texttt{arm-rm,gf}}{=}} 1.$$

[50] /T/prob/P/margzimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}) = 0 \overset{\texttt{mz}}{\implies} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

Proof.

We have that

$$0 \overset{\texttt{mz}}{=} \Pr(\mathrm{X} = \mathrm{x}) \overset{[\ldots]^{47}_{\mathrm{P}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}'),$$

so in particular

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

[47] /T/prob/P/mprobg

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

Proof.

We have by definition that for all $\mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{46}_{\mathrm{D}}}{=} \sum_{?\in\mathbf{?}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?),$$

mm

so

$$\Pr(\mathrm{X} = \mathrm{x}) \overset{[\ldots]^{46}_{\mathrm{D}}}{=} \sum_{\mathrm{y}\in\mathbf{Y}} \sum_{?\in\mathbf{?}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?) \overset{\text{mm}}{=} \sum_{\mathrm{y}\in\mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[51] /T/prob/P/marggimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \overset{\texttt{mg}}{\implies} \Pr(\mathrm{X} = \mathrm{x}) > 0.$$

Proof.

We have that

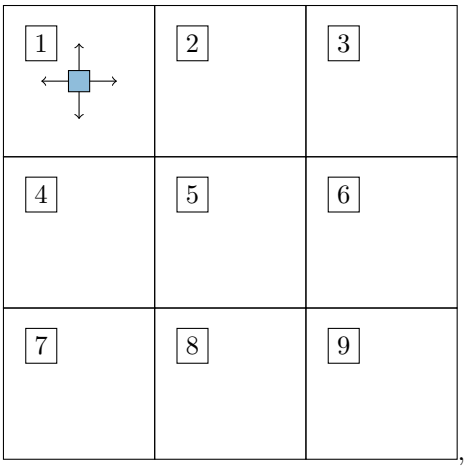
$$\Pr(\mathrm{X} = \mathrm{x}) \overset{[\cdots]^{47}_{\mathrm{P}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}') \geq \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \overset{\texttt{mg}}{>} 0.$$

Reinforcement Learning Theory

Adapted from *Neuro-Dynamic Programming* by
Dmitri P. Bertsekas and John N. Tsitsiklis.

Physical intuition.

Consider the following "environment":



,

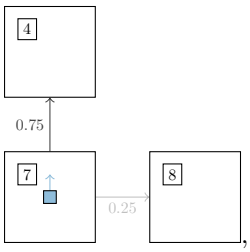
where there are a total of 9 "states" (▣), an "agent" (■), and a set of "actions" (→) that the agent can take at any state.

At every "step" in time, the agent is in one of the states, and must take one of the actions.

At any step, if the agent takes an action in a state, it will be in a state at the next step according to a "transition distribution" (that only depends on the current state and action, not the current step).

For example, the agent could

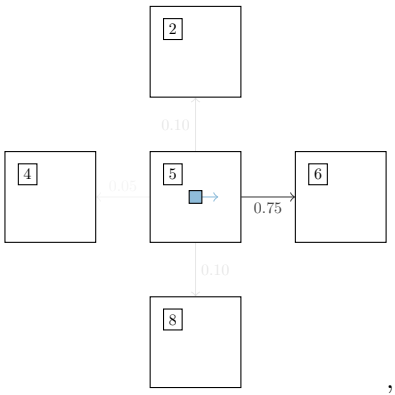
- take action ↑ at state 7:



,

  and

- take action → at state 5:



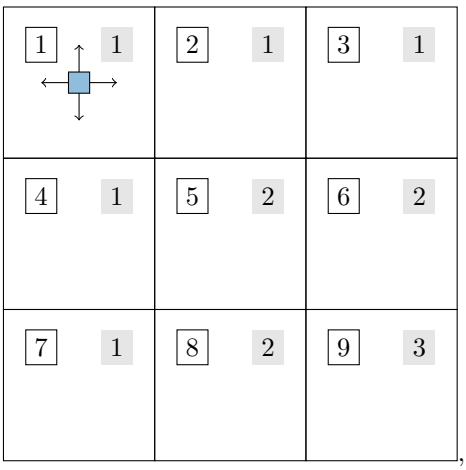
,

showing the conditional probability of being in another state at the next step according to the transition distribution.

However, this begs the question, what state is the agent at initially (step 1)? To decide this, there is also an initial distribution over the states.

Now, let's make things a bit more interesting.

Let every state also have a return that the agent recieves from it, leaving our environment looking something like:

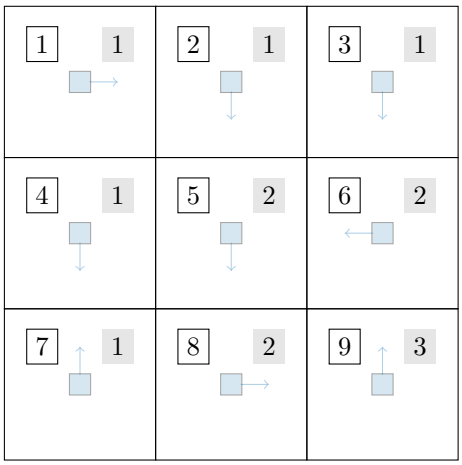


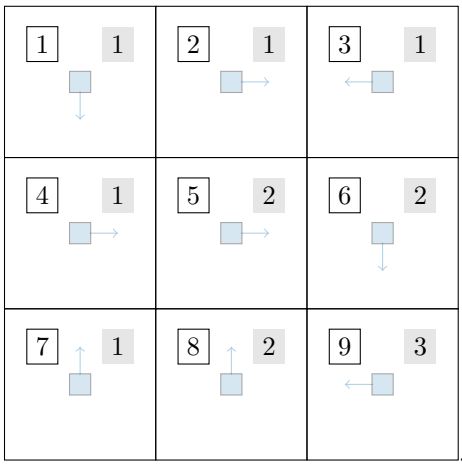
,

with the returns (·) shown.

Also, let there be some fixed "step limit" such that, when the current step reaches it, everything ends.

Now, suppose you can "program" the agent as follows: for every state and step, you can fix the action that the agent takes. For example, here's what you might program the agent to do at every state for two different steps:





.

We'll call each such programming a "strategy".

We know that the agent starts in some state according to the initial distribution. After this, we follow the strategy, transitioning states according to the transition distribution, recieving returns at every state until we reach the step limit. Let's refer to this process as a "run", and the sum of the returns over all a run's states as the "total return".

Because there is some initial distribution and transition distribution that are fixed over runs, we know there must be some average total return for every strategy.

Now, consider the following question: what strategy has the greatest average total return?

Notation.

To answer this question, let's establish some notation.

[Define]$^{26}_{\mathrm{D}}$

$$\mathbf{S} \coloneqq \text{the set of states},$$

where $\mathbf{S}$ is finite and "s" denotes an element of $\mathbf{S}$.

[Define]$^{30}_{\mathrm{D}}$

$$\mathbf{A} \coloneqq \text{the set of actions},$$

where $\mathbf{A}$ is finite and "a" denotes an element of $\mathbf{A}$.

[Define]$^{19}_{\mathrm{D}}$

$$[R(\cdot) : \mathbf{S} \to \mathbb{R}]^{17}_{\mathrm{N}} \coloneqq \text{mapping from states to returns}.$$

[Define]$^{21}_{\mathrm{D}}$

$$T \coloneqq \text{the step limit, where } T \geq 1,$$

where t denotes a step, with $1 \leq t \leq T$.

[Define]$^{34}_{\mathrm{D}}$

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies},$$

where π denotes an element of $\mathbf{\Pi}$, with

$$\pi = (\pi_1(\cdot), \ldots, \pi_{T-1}(\cdot)) \coloneqq \text{a strategy, where each "step-strategy" } \pi_t(\cdot) : \mathbf{S} \to \mathbf{A}.$$

[Note that]$^{4}_{\mathrm{P}}$ the total number of strategies can be written as

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|.$$

[Define]$^{23}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}$,

$$[\mathrm{P_I}(\mathrm{s})]^{13}_{\mathrm{N}} \coloneqq \text{the initial distribution probability of starting in state s.}$$

[Define]$^{28}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$,

$$\mathrm{P_A}(\pi, t : \mathrm{a} \mid \mathrm{s}) \coloneqq \begin{cases} 1 & \pi_t(\mathrm{s}) = \mathrm{a} \\ 0 & \pi_t(\mathrm{s}) \neq \mathrm{a}. \end{cases}$$

[Define]$^{31}_{\mathrm{D}}$ for any $\mathrm{s}, \mathrm{s}' \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$,

$$\mathrm{P_T}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}') \coloneqq \text{the transition distribution probability of going to state s after action a in state s}'.$$

[Define]$^{29}_{\mathrm{D}}$ for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{variable for the state at step } t,$$

with domain $\mathbf{S}$.

[Define]$^{27}_{\mathrm{D}}$ for all $1 \leq t \leq T-1$,

$$\mathrm{A}_t \coloneqq \text{variable for the action at step } t,$$

with domain $\mathbf{A}$.

[Let]$^{49}_{\mathrm{D}}$

$$\mathrm{S} \leftarrow \{(\mathrm{S}_1, \mathbf{S}), (\mathrm{A}_1, \mathbf{A}), (\mathrm{S}_2, \mathbf{S}), \ldots, (\mathrm{A}_{T-1}, \mathbf{A}), (\mathrm{S}_T, \mathbf{S})\},$$

and, for each $\pi \in \mathbf{\Pi}$, let $\mathrm{D}_\pi \in \mathbf{D}$ be a distribution.

[Let]$^{20}_{\mathrm{D}}$

$$\mathrm{Pr}_\pi(\cdot) \coloneqq \mathrm{Pr}(\cdot),$$

where $\mathrm{D} = \mathrm{D}_\pi$, as applied to [joint probabilities]$^{43}_{\mathrm{D}}$, [marginal probabilities]$^{46}_{\mathrm{D}}$, and [conditional probabilities]$^{48}_{\mathrm{D}}$.

Because we follow a fixed strategy, [we know that]$^{25}_{\mathrm{D}}$, for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$, $1 \leq t \leq T - 1$, when defined,

$$\Pr_\pi (\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_\mathrm{A} (\pi, t : \mathrm{a} \mid \mathrm{s}),$$

[from which it follows]$^{3}_{\mathrm{P}}$ that A_t is independent of all previous states and actions given S_t. Notationally, for all $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T - 1$, when defined,

$$\Pr_\pi (\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi (\mathrm{A}_t \mid \mathrm{S}_t).$$

[We know that]$^{18}_{\mathrm{D}}$ for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$,

$$\Pr_\pi (\mathrm{S}_1 = \mathrm{s}) = \mathrm{P}_\mathrm{I} (\mathrm{s}),$$

and [we know that]$^{33}_{\mathrm{D}}$ for all $\pi \in \mathbf{\Pi}$, $\mathrm{s}, \mathrm{s}' \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$, $2 \leq t \leq T$, when defined,

$$\Pr_\pi (\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') = \mathrm{P}_\mathrm{T} (\mathrm{s} \mid \mathrm{a}, \mathrm{s}'),$$

[where]$^{24}_{\mathrm{D}}$ S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for all $\pi \in \mathbf{\Pi}$, $2 \leq t \leq T$, when defined,

$$\Pr_\pi (\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi (\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}).$$

Propositions.

Let's now describe some preliminary results using this notation.

One of the first questions we may ask is, does our notation completely describe the distribution of runs for a particular strategy? That is,

$$\text{for any } \mathrm{s}_1, \ldots, \mathrm{s}_T \in \mathbf{S}, \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{A},$$

and

$$\text{for any strategy } \pi \in \mathbf{\Pi},$$

can we determine the value of

$$\mathrm{Pr}_\pi\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T\right)?$$

The answer to this question is: yes! We have all the information we need to completely describe this joint distribution. [It can be written][5P] as

$$\begin{aligned}\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right)\times & \\ \mathrm{P}_\mathrm{A}\left(\pi, 1 : \mathrm{a}_1 \mid \mathrm{s}_1\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_2 \mid \mathrm{a}_1, \mathrm{s}_1\right)\times & \\ \mathrm{P}_\mathrm{A}\left(\pi, 2 : \mathrm{a}_2 \mid \mathrm{s}_2\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_3 \mid \mathrm{a}_2, \mathrm{s}_2\right)\times & \\ \ldots\times & \\ \mathrm{P}_\mathrm{A}\left(\pi, T-1 : \mathrm{a}_{T-1} \mid \mathrm{s}_{T-1}\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_T \mid \mathrm{a}_{T-1}, \mathrm{s}_{T-1}\right), & \end{aligned}$$

or, in more compact [product notation][14N],

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t=1}^{T-1} \mathrm{P}_\mathrm{A}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right)\mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

---

[41] /T/prob/D/seqp

Definition.

We let for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}, (\exists\ 1 \leq t < \infty : \mathrm{SQI}\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1) \implies \Pr\,(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0\},$$

that is, the set of all probability sequences, where "sqp" denotes an arbitrary such probability sequence.

[46] /T/prob/D/mprob

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}),$$

that is, a marginal probability.

[42] /T/prob/D/dist

Definition.

We define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\ \mathrm{d}, \mathrm{d}' \in \mathrm{D}, \forall\ \mathrm{x} \in \mathbf{X}, \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions, where "D" denotes an abitrary such distribution, and "d" denotes an arbitrary sequence in D.

[57] /T/prob/D/asn

Definition.

We let for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$] and all $\mathrm{x} \in \mathbf{X}$,

$$\mathrm{X} = \mathrm{x}$$

denote a joint assignment (assignment for short) [or, marginal assignment (in simple English, a list of equalities that doesn't cover the entire support is a marginal assignment)].

If a joint assignment or marginal assignment was previously specified, we may later drop "X =", for "implied assignment":

$$\mathrm{x}.$$

If a joint assignment or marginal assignment was never specified, we assume an arbitrary $\mathrm{x} \in \mathbf{X}$ for "implied generality":

$$\mathrm{X}.$$

For contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}'), (\mathrm{Y}, \mathbf{Y}), \ldots, (\mathrm{Y}', \mathbf{Y}')\}$] and all $\mathrm{x} \in \mathbf{X}, \ldots, \mathrm{x}' \in \mathbf{X}'$, This expands as

$$\mathrm{X} = \mathrm{x}, \ldots, \mathrm{X}' = \mathrm{x}',$$

with implied assignment as

$$\mathrm{x}, \ldots, \mathrm{x}',$$

and with implied generality as

$$\mathrm{X}, \ldots, \mathrm{X}'.$$

[43] /T/prob/D/prob

Definition.

We define for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$ and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a joint probability.

[38] /T/prob/D/seq

Definition.

We let, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQ} \text{ denote the set of all sequences,}$$

where "sq" denotes an arbitrary such sequence.

Let $1 \leq t < \infty$ denote an arbitrary trial.

Because sequences are infinite (and cannot be written out), we use the notation

$$\mathrm{SQI}\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) \coloneqq \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$.

[56] /T/prob/D/pin

Definition.

We let, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, the notation

$$\mathrm{x} \in \mathbf{X}$$

indicate an arbitrary x in $\mathbf{X}$.

For contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}'), (?, \mathbf{?})\}$, this expands as

$$\mathrm{x} \in \mathbf{X}, \ldots, \mathrm{x}' \in \mathbf{X}'$$

to indicate an arbitrary combination of x in $\mathbf{X}$, …, x' in $\mathbf{X}'$.

[40] /T/prob/D/prsq

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \mathrm{SQI}(\mathrm{sq}, t', \mathrm{X} = \mathrm{x})}{t} \right)$$

if such a limit exists, and undefined otherwise; that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[36] /T/prob/D/xyz

Definition.

We have by definition that

$$\mathrm{X} \coloneqq \text{an arbitrary variable with domain } \mathbf{X},$$

where we use "x" to denote an element of $\mathbf{X}$.

We implicitly maintain that anything we describe using X and $\mathbf{X}$ also applies when these are replaced with one or more variables/domains: $\mathrm{X}/\mathbf{X}, \ldots, \mathrm{X}'/\mathbf{X}'$. Notational expansions are described where applicable.

Define $\mathrm{Y}, \mathbf{Y}, \mathrm{y}$, and $\mathrm{Z}, \mathbf{Z}, \mathrm{z}$ similarly.

Define $?, \mathbf{?}, ?$ similarly, but instead implicitly maintain that anything we describe using ? and **?** also applies when these are replaced with *zero* or more variables/domains.

[37] /T/prob/D/spt

Definition.

We let the notation

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\} \text{ establish a support consisting of variable } \mathrm{X} \text{ with domain } \mathbf{X},$$

which expands as

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), \ldots, (\mathrm{X}', \mathbf{X}')\}.$$

[48] /T/prob/D/cprob

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \frac{\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{Y} = \mathrm{y})},$$

and undefined when $\Pr(\mathrm{Y} = \mathrm{y}) = 0$; that is, a conditional probability.

[3] /T/prob/T/rl/P/actcind

Proposition.

We have that A_t is independent of all previous states and actions given S_t. Notationally, for all $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$, when defined,

$$\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t).$$

Proof.

Because $\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1)$ is defined, we have that

$$\Pr_\pi(\mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) > 0,$$

implying that

$$\Pr_\pi(\mathrm{S}_t) > 0.$$

Therefore, $\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t)$ is also defined. Suppose $\mathrm{A}_t = \mathrm{a} \in \mathbf{A}$ and $\mathrm{S}_t = \mathrm{s} \in \mathbf{S}$. [It follows by definition]$^{25}_{\mathrm{D}}$ that

$$\Pr_\pi(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_\mathrm{A}(\pi, t : \mathrm{a} \mid \mathrm{s}),$$

so [there are two cases]$^{28}_{\mathrm{D}}$ to consider:

- Case 1: $\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t) = 1$. Then, po

$$\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \overset{\text{(po)->[...]}^{52}_{\mathrm{P}}\ \text{po}}{=} 1 = \Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t).$$

- Case 2: $\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t) = 0$. Then, pz

$$\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \overset{\text{(pz)->[...]}^{53}_{\mathrm{P}}\ \text{pz}}{=} 0 = \Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t).$$

[4] /T/prob/T/rl/P/npol

Proposition.

We have that the total number of strategies can be written as

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|.$$

Proof.

This follows from simple combinatorics. Consider an arbitrary strategy

$$\pi = (\pi_1(\cdot), \ldots, \pi_T(\cdot)).$$

Because each step-strategy $\pi_t(\cdot) : \mathbf{S} \to \mathbf{A}$, there are a total of $|\mathbf{S}| \times |\mathbf{A}|$ distinct $\pi_t(\cdot)$ possible.

Because each strategy $\pi \in \mathbf{\Pi}$ consists of T such step-strategies $\pi_t(\cdot)$, there are a total of

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|$$

distinct π possible.

[5] /T/prob/T/rl/P/pexpr

Proposition.

We have that for

$$\text{any } \mathrm{s}_1, \ldots, \mathrm{s}_T \in \mathbf{S}, \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi \left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T\right)$$

as

$$\mathrm{Pr}_\pi \left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_\mathrm{I} \left(\mathrm{s}_1\right) \times \prod_{t=1}^{T-1} \mathrm{P}_\mathrm{A} \left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right) \mathrm{P}_\mathrm{T} \left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

Proof.

[It can be shown inductively]$^{11}_{\mathrm{P}}$ that for any $1 \leq t \leq T$, for

$$\text{any } \mathrm{s}_1, \ldots, \mathrm{s}_t \in \mathbf{S}, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi \left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t\right)$$

as

$$\mathrm{Pr}_\pi \left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = \mathrm{P}_\mathrm{I} \left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A} \left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T} \left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right),$$

for which the result follows in particular when $t = T$.

[13] /T/prob/T/rl/N/paramprob

Note.

We use the notation $\Pr(\cdot)$ and $\mathrm{P}(\cdot)$ to distinguish between a probability and a "probability parameterization", respectively.

A probability parameterization is simply a function mapping to probability values, and is defined for every possible input. The form of the inputs are implicit in its definition.

A probability, however, may or may not be defined, specifically when it is a conditional probability.

When we correspond a probability with a probability parameterization, we mean to say that, for a particular input, they are equal when the probability is defined.

For example, suppose $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ with any D.

For all $\mathrm{x} \in \mathbf{X}, \mathrm{y} \in \mathbf{Y}$, if we correspond $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ with $\mathrm{P}(\mathrm{x} \mid \mathrm{y})$, we mean to say that when $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ is defined (that is, when $\Pr(\mathrm{Y} = \mathrm{y}) > 0$),

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \mathrm{P}(\mathrm{x} \mid \mathrm{y}).$$

[14] /T/prob/T/rl/N/prodnot

Note.

Consider an arbitrary function $f(\cdot)$. The notation

$$\prod_{x=a}^{b} f(x),$$

where a, b are integers, denotes the product $f(a) \times f(a+1) \times \cdots \times f(b-1) \times f(b)$ when $a \leq b$, and 1 otherwise. That is,

$$\prod_{x=a}^{b} f(x) = \begin{cases} f(a) \times f(a+1) \times \cdots \times f(b-1) \times f(b) & a \leq b \\ 1 & a > b. \end{cases}$$

[17] /T/prob/T/rl/N/fnnot

Note.

We use the notation

$$f(\cdot) : X \to Y$$

to denote a function with domain X and range Y; that is,

$$\text{for all } x \in X, \text{ there is some } y \in Y \text{ such that } f(x) = y.$$

[18] /T/prob/T/rl/D/ind

Definition.

We have by definition that for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$,

$$\mathrm{Pr}_{\pi}\left(\mathrm{S}_1 = \mathrm{s}\right) = \mathrm{P}_{\mathrm{I}}\left(\mathrm{s}\right).$$

[19] /T/prob/T/rl/D/rewards

Definition.

We have by definition that

$$R(\cdot) : \mathbf{S} \to \mathbb{R} \coloneqq \text{mapping from states to returns.}$$

[20] /T/prob/T/rl/D/pstr

Definition.

We have by definition that

$$\Pr_\pi(\cdot) \coloneqq \Pr(\cdot),$$

where $\mathrm{D} = \mathrm{D}_\pi$, as applied to [joint probabilities]$^{43}_{\mathrm{D}}$, [marginal probabilities]$^{46}_{\mathrm{D}}$, and [conditional probabilities]$^{48}_{\mathrm{D}}$.

[21] /T/prob/T/rl/D/steplim

Definition.

We have by definition that

$$T \coloneqq \text{the step limit, where } T \geq 1,$$

where t denotes a step, with $1 \leq t \leq T$.

[23] /T/prob/T/rl/D/inddef

Definition.

We have by definition that for any $s \in \mathbf{S}$,

$$P_I(s) \coloneqq \text{the initial distribution probability of starting in state } s.$$

[24] /T/prob/T/rl/D/trdcind

Definition.

We have by definition that S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for all $\pi \in \mathbf{\Pi}$, $2 \leq t \leq T$, when defined,

$$\Pr_\pi(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}).$$

[25] /T/prob/T/rl/D/pact

Definition.

We have by definition that, for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$, $1 \leq t \leq T-1$, when defined,

$$\mathrm{Pr}_{\pi}\left(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}\right) = \mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right).$$

[26] /T/prob/T/rl/D/states

Definition.

We have by definition that

$$\mathbf{S} := \text{the set of states},$$

where $\mathbf{S}$ is finite and "s" denotes an element of $\mathbf{S}$.

[27] /T/prob/T/rl/D/rvat

Definition.

We have by definition that for all $1 \leq t \leq T - 1$,

$$\mathrm{A}_t \coloneqq \text{variable for the action at step } t,$$

with domain $\mathbf{A}$.

[28] /T/prob/T/rl/D/pactdef

Definition.

We have by definition that for any $\mathrm{s} \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$,

$$\mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right) \coloneqq \begin{cases} 1 & \pi_t\left(\mathrm{s}\right) = \mathrm{a} \\ 0 & \pi_t\left(\mathrm{s}\right) \neq \mathrm{a}. \end{cases}$$

[29] /T/prob/T/rl/D/rvst

Definition.

We have by definition that for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{variable for the state at step } t,$$

with domain $\mathbf{S}$.

[30] /T/prob/T/rl/D/actions

Definition.

We have by definition that

$$\mathbf{A} := \text{the set of actions},$$

where $\mathbf{A}$ is finite and "a" denotes an element of $\mathbf{A}$.

[49] /T/prob/T/rl/D/psp

Definition.

We have by definition that

$$\mathrm{S} \leftarrow \{(\mathrm{S}_1, \mathbf{S}), (\mathrm{A}_1, \mathbf{A}), (\mathrm{S}_2, \mathbf{S}), \ldots, (\mathrm{A}_{T-1}, \mathbf{A}), (\mathrm{S}_T, \mathbf{S})\},$$

and, for each $\pi \in \mathbf{\Pi}$, let $\mathrm{D}_\pi \in \mathbf{D}$ be a distribution.

[31] /T/prob/T/rl/D/trddef

Definition.

We have by definition that for any $\mathrm{s}, \mathrm{s}' \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$,

$\mathrm{P}_{\mathrm{T}}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}') :=$ the transition distribution probability of going to state s after action a in state s'.

[33] /T/prob/T/rl/D/trd

Definition.

We have by definition that for all $\pi \in \mathbf{\Pi}$, $\mathrm{s}, \mathrm{s}' \in \mathbf{S}, \mathrm{a} \in \mathbf{A}$, $2 \leq t \leq T$, when defined,

$$\mathrm{Pr}_{\pi}\left(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}'\right) = \mathrm{P}_{\mathrm{T}}\left(\mathrm{s} \mid \mathrm{a}, \mathrm{s}'\right).$$

[34] /T/prob/T/rl/D/strs

Definition.

We have by definition that

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies},$$

where π denotes an element of $\mathbf{\Pi}$, with

$$\pi = (\pi_1(\cdot), \ldots, \pi_{T-1}(\cdot)) \coloneqq \text{a strategy, where each "step-strategy" } \pi_t(\cdot) : \mathbf{S} \to \mathbf{A}.$$

[11] /T/prob/T/rl/P/pexpr/P/pexprt

Proposition.

We have that for any $1 \leq t \leq T$, for

$$\text{any } \mathrm{s}_1, \ldots, \mathrm{s}_t \in \mathbf{S}, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t\right)$$

as

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Proof.

Base case. When $t = 1$, we have [by definition][18]$_\mathrm{D}$ that $\mathrm{Pr}_\pi\left(\mathrm{s}_1\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right)$.

Inductive hypothesis. Assume that for $t - 1$ such that $1 \leq t - 1 \leq T - 1$, the proposition holds.
tmg tml

That is,

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \underset{\text{ih}}{=} \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-2} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Inductive step. We want to show that the proposition holds for t.

There are two cases to consider:

- Case 1: $\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \underset{\text{co}}{=} 0$. Then,

  (co)->[...][50]$_\mathrm{P}$

  $$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) \underset{\text{coz}}{=} 0,$$

  and

  $$\begin{aligned}
  &\mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \\
  &= \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-2} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \times \\
  &\quad \mathrm{P}_\mathrm{A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
  &\underset{\text{ih}}{=} \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
  &\underset{\text{co coz}}{=} 0 = \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right).
  \end{aligned}$$

– Case 2: $\Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) > 0$. Then,
ct

$$\overset{\texttt{(ct)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \text{ is defined}}, \overset{\texttt{((ct)->}[\ldots]^{51}_{\mathrm{P}}\texttt{)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(a_{t-1} \mid s_{t-1}) \text{ is defined}},$$
daf da

and we have

$$\overset{\texttt{(daf,tmg,tml)->}[\ldots]^{3}_{\mathrm{P}}}{\Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) = \Pr_\pi(a_{t-1} \mid s_{t-1})} \overset{\texttt{(da,tmg,tml)->}[\ldots]^{25}_{\mathrm{D}}}{= \mathrm{P_A}(\pi, t-1 : a_{t-1} \mid s_{t-1})}.$$
cia ad

[There are two cases]$^{28}_{\mathrm{D}}$ to consider:

– Case 2.1: $\mathrm{P_A}(\pi, t-1 : a_{t-1} \mid s_{t-1}) = 1$. Then,
cto

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}, a_{t-1}) &\overset{\substack{\texttt{daf}\\ \texttt{md}}}{=} \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia-ad}}{=} \mathrm{P_A}(\pi, t-1 : a_{t-1} \mid s_{t-1}) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cto}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\substack{\texttt{ct}\\ \texttt{mg}}}{>} 0.
\end{aligned}$$

Therefore,

$$\overset{\texttt{(md-mg)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \text{ is defined}}, \overset{\texttt{((md-mg)->}[\ldots]^{51}_{\mathrm{P}}\texttt{)->}[\ldots]^{48}_{\mathrm{D}}}{\Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \text{ is defined}}$$
dsf ds

and we have

$$\overset{\texttt{(dsf,tmg,tml)->}[\ldots]^{24}_{\mathrm{D}}}{\Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) = \Pr_\pi(s_t \mid a_{t-1}, s_{t-1})} \overset{\texttt{(ds,tmg,tml)->}[\ldots]^{33}_{\mathrm{D}}}{= \mathrm{P_T}(s_t \mid a_{t-1}, s_{t-1})}.$$
cis sd

Therefore,

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_t) &\overset{\texttt{daf,dsf}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \\
&\qquad \Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia,cis}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(a_{t-1} \mid s_{t-1}) \times \Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \\
&\overset{\texttt{ih,ad,sd}}{=} \mathrm{P_I}(s_1) \times \prod_{t'=1}^{t-1} \mathrm{P_A}(\pi, t' : a_{t'} \mid s_{t'}) \mathrm{P_T}(s_{t'+1} \mid a_{t'}, s_{t'}).
\end{aligned}$$

– Case 2.2: $\mathrm{P_A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) = 0$. Then, ctt

daf

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}, \mathrm{a}_{t-1}\right) = \mathrm{Pr}_\pi\left(\mathrm{a}_{t-1} \mid \mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \times \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right)$$

afd

cia-ad

$$= \mathrm{P_A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right)$$

ctt

$$= 0.$$

afz

Therefore,

(afd-afz)->[...]50 P

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = 0,$$

cttz

so

cttz ctt

$$\mathrm{Pr}_\pi\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = 0 = \mathrm{P_I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$